

Matrix inequalities and norms

Bapat's q -permanent monotonicity conjecture

Difficulty: extreme **Importance:** interesting to specialist**Status:** Partially resolved

Rating rationale: Strict monotonicity on the whole PSD cone is a longstanding generalized-permanent barrier; its immediate impact is in specialist matrix-function inequalities.

Problem statement

For a permutation $\sigma \in S_n$, let

$$\text{inv}(\sigma) = \#\{(i, j) : 1 \leq i < j \leq n, \sigma(i) > \sigma(j)\}.$$

For real q and $A = (a_{ij}) \in \mathbb{C}^{n \times n}$, define

$$P_q(A) = \sum_{\sigma \in S_n} q^{\text{inv}(\sigma)} \prod_{i=1}^n a_{i, \sigma(i)}, \quad 0^0 = 1.$$

Is it true that, for every $n \geq 2$, every non-diagonal Hermitian PSD matrix A with $a_{ii} > 0$ for all i , and every $-1 \leq q_1 < q_2 \leq 1$,

$$P_{q_1}(A) < P_{q_2}(A)?$$

The expression is real for Hermitian A . The order of rows and columns in the inversion statistic is fixed.

Relevance and ratings

This polynomial interpolates $\det A$ at $q = -1$, the diagonal product at $q = 0$, and $\text{per } A$ at $q = 1$. The conjecture would compare a continuous family of matrix functions on the PSD cone, extending determinant/permanent inequalities. The non-diagonal and positive-diagonal hypotheses remove constant-polynomial cases.

References

- R. B. Bapat and A. K. Lal, *Inequalities for the q -permanent*, Linear Algebra and its Applications 197–198 (1994), 397–409, monotonicity conjecture ([original paper](#)).
- L. Mitchell, *A note on Bapat's q -permanent conjecture*, Operators and Matrices 14 (2020), 915–919, un-numbered Conjecture on p.917 and Theorems 1–3 ([primary paper](#)).
- C. M. da Fonseca, *The μ -permanent revisited* (2018), §4, related conjectures and corrections concerning ordered graph cases ([primary manuscript](#)).

Status check — 2026-09-10

Mitchell proves that the original positive-definite conjecture is equivalent to the displayed extension, and proves rank-one and order-three cases. Searches for “Bapat q -permanent conjecture”, “monotonicity”, “proof”, “counterexample”, and 2025–2026 found no resolution. The stronger proposed extension beyond $[-1, 1]$ is not included; Mitchell explains why extending that assertion to all PSD matrices fails. No recent explicit reaffirmation of the exact conjecture was found.

Independent audit: Independently rechecked Mitchell's PSD-extension conjecture and rank-one theorem, and searched for later q -permanent monotonicity resolutions. Rank-one and low-order cases are substantive parts of the displayed family; the general case still has only historical-source status evidence.